

Rishikesh Kumar

ML Engineer | Data Scientist

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Education

University	Degree	Branch	Year
IIT Bombay	M.Tech	IEOR	Jul 2024 – Jul 2026
Guru Ghasidas University	B.Tech	CSE	Jul 2020 – Jul 2024

Internships

Full-Stack Developer Intern | Vajra Labs, IIT Bombay | Powai, Maharashtra **Aug 2025 – Present**

- Contributed to **2 projects**, developing the **frontend of the company website** and a **sandbox-based malware analysis platform**
- Led a **3-member team**, delivering the project on schedule by coordinating tasks and implementing Agile methodologies

Web Developer Intern | RMgX Technologies | Gurugram, Haryana **Jun 2022 – Aug 2022**

- Created and integrated **10+ fully responsive pages** for the **company website** using **Next.js**
- Collaborated with the **UI/UX team** to translate design mockups into frontend code, ensuring a cohesive and intuitive experience

Projects

Explainable Depression Detection from Social Media | Course Project : DL NLP | Prof. Pushpak Bhattacharyya **Aug 2025 – Nov 2025**

- Built an **NLP pipeline** for **depression detection** from social media posts, using **double domain adaptation** and **guided masking**
- Designed an **explainability pipeline** by combining **Integrated Gradients** with an **LLM-based teacher-student distillation method**

- Achieved an **F1 score of 0.78** on test data, and extracted top-k words contributing most strongly to positive depression detection

HeadBall2 Game Player | Self Project | Real-Time Mobile Game Automation **Feb 2025 – Mar 2025**

- Developed a **real-time computer vision** pipeline for mobile game automation using **OpenCV**, from live video streams at **30 FPS**
- Designed a **reinforcement learning system** with **WebSocket-based** client-server communication, achieving **50 ms** end-to-end latency
- Built a **mobile automation framework** integrating **ADB**, **v4l2** and **Python**, enabling reliable device control and screen streaming

Research & Publication

Classification of Plant Diseases using SOTA Deep learning Algorithms **July 2024**

International Journal of Computing | DOI: 10.47839/ijc.23.4.3542

- Implemented and evaluated multiple **CNN architectures** on a dataset of **87,900+ leaf images** across **38 classes**
- Achieved up to **97.31% validation accuracy**, showing robust generalization across multiple plant diseases

MTP Thesis & Seminar

Parameter Specific Literature Retrieval | Master's Thesis Project (MTP) | Prof. Balamurugan Palaniappan **Jul 2025 – Nov 2025**

- Streamlined a **biomedical literature retrieval pipeline** to search PubMed data and extract **parameter-specific** clinical evidence
- Improved **entity extraction F1-score to 90%** by fine-tuning a **PubMedBERT-based NER model** on a manually annotated dataset
- Converted unstructured NER output into structured XML using a **BART-based Seq2Tree model** for parameter range extraction
- Implemented a **distributed ML pipeline** with **Ray**, **PyTorch**, and **HuggingFace Transformers**; deployed it with **Next.js/FastAPI**

Multimodal Transformer Architecture and Optimization | Seminar | Prof. Urban Larsson **Jan 2025 – Apr 2025**

- Implemented the **Attention is All You Need** paper in **PyTorch** and presented an in-depth analysis of the paper during the seminar
- Analyzed in-depth concepts of **An Image is Worth 16x16 Words** and presented how to use images in **transformer architecture**
- Conducted a seminar presentation on the research behind the **LoRA** paper and discussed its advantages and disadvantages

Position of Responsibility

Teaching Assistant | IE 643, Deep Learning – Theory and Practice **Jul 2025 – Nov 2025**

- Organized and led **weekly programming sessions** to reinforce core concepts and enhance **practical skills** in DL for **80+ students**
- Managed and **graded** student copies and projects, providing detailed **feedback** to improve their understanding of complex topics
- Facilitated **project planning** for the student teams and conducted **regular doubt sessions** ensuring project success and completion

Website Coordinator | IEOR Day, IIT Bombay **Jan 2025 – Mar 2025**

- Organized and coordinated** the annual **IEOR Day** with a **team of 30+ members** to enhance event visibility and resource access
- Developed and maintained the **event's official website**, providing attendees with event schedules, announcements and updates

Technical Skills

- Languages:** Python, C, C++, JavaScript, SQL, HTML, CSS, Java, Lua, TypeScript
- Frameworks & Libraries:** HuggingFace Transformers, PyTorch, Scikit-Learn, Django, React.js, Tailwind CSS
- Systems & Tools:** PostgreSQL, Redis, Docker, Docker Compose, Git, RabbitMQ, Celery, Ray, WebSockets, Linux, REST APIs

Achievements

- Achieved **99.30 percentile (2024)** and **99.13 percentile (2023)** in **GATE CSE**
- Secured **World Rank 32** in **CodeChef Starters 42**
- Secured **Rank 1** in **HackForCG Hackathon** by developing an **ML-based computer vision** solution using **OpenCV** and **Haar Cascade**